

Practical No. 03
ZOO 1203 - General Entomology
Different mouthparts of insects

Objectives:

After completing the practical you should be able to;

- Dissect mouthparts of an insect
 - Identify different mouthparts of an insect
 - Predict the feeding mode of an insect by looking at the mouthparts
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The ancestors of present-day insects were probably worm-like arthropods with a simple mouth opening near the front of a bilaterally symmetrical body. Appendages near the mouth opening came to be adapted for gathering and manipulating bits of solid food. As insects evolved, they became more complex, expanded in range, and adapted to new food resources. The structure and function of their mouthparts changed right along with their evolving diet and life style.

Different mouth parts commonly seen in insects are:

- Chewing mouthparts (Ex: Cockroach, Grasshopper)
- Piercing and sucking (Ex: Mosquito, Paddy bug)
- Siphoning mouthparts (Ex: Butterfly)
- Sponging mouthparts (Ex: Housefly)
- Chewing and lapping (Ex: Honey bee)

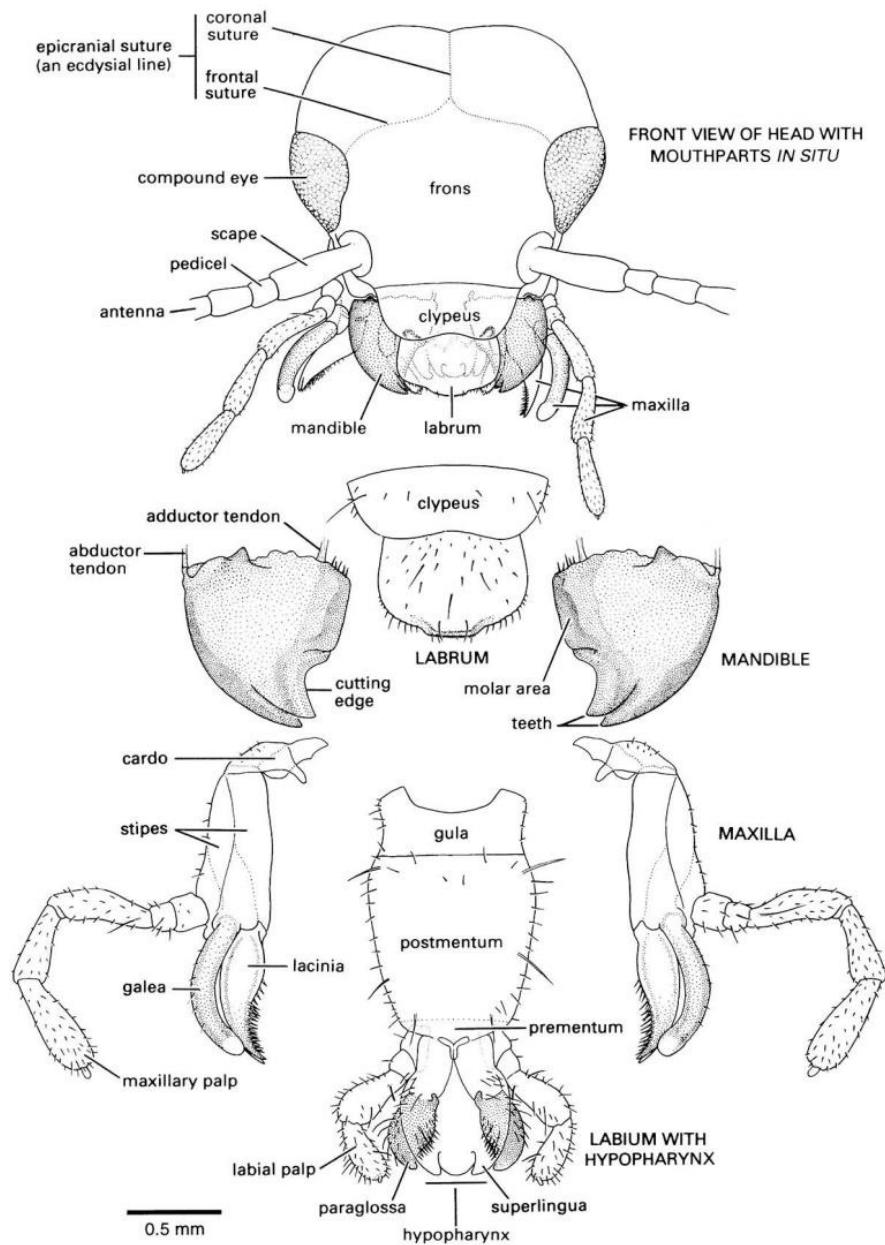
CHEWING MOUTHPARTS

This type of mouth parts are supposed to be the most primitive type as the other types are believed to be evolved from biting and chewing type of mouth parts.

These consist of:

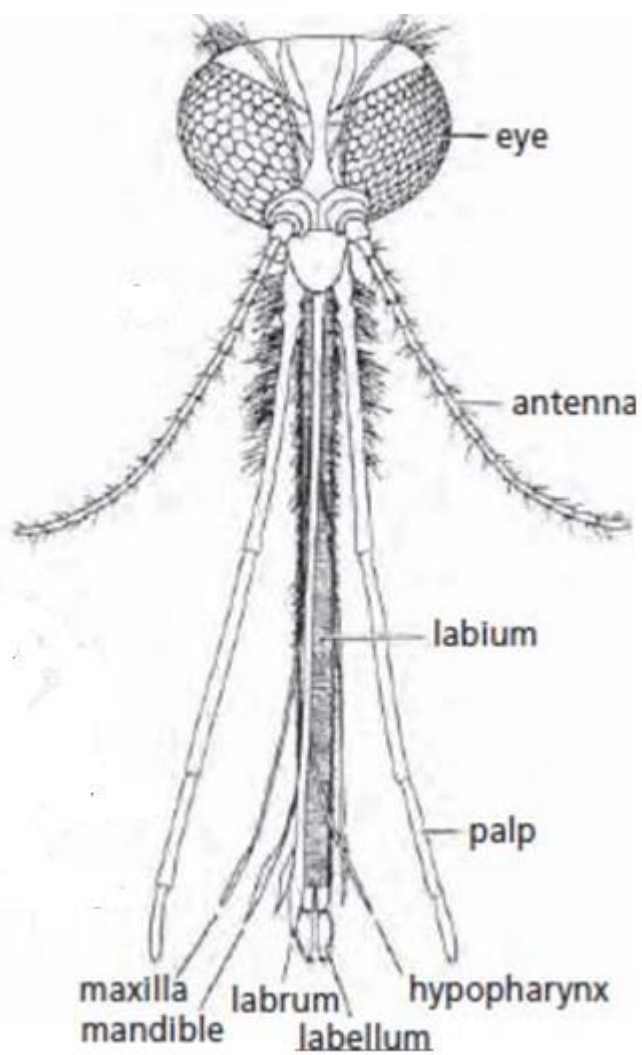
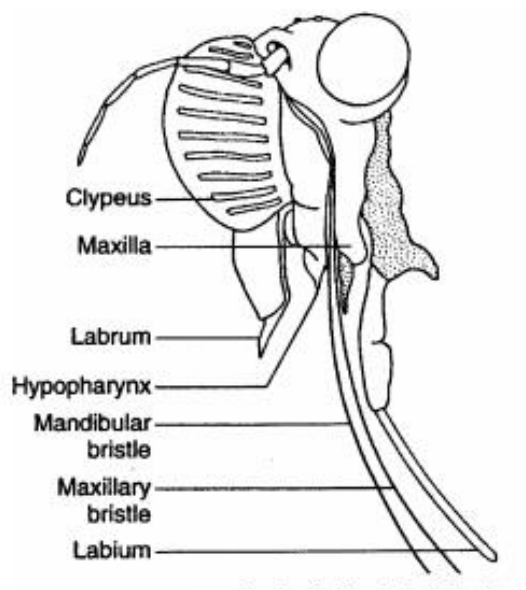
1. Labrum forming upper lip
2. Mandibles

3. First maxillae
4. Second maxillae forming lower lip
5. Hypo pharynx
6. Epipharynx.



PIERCING AND SUCKING MOUTHPARTS

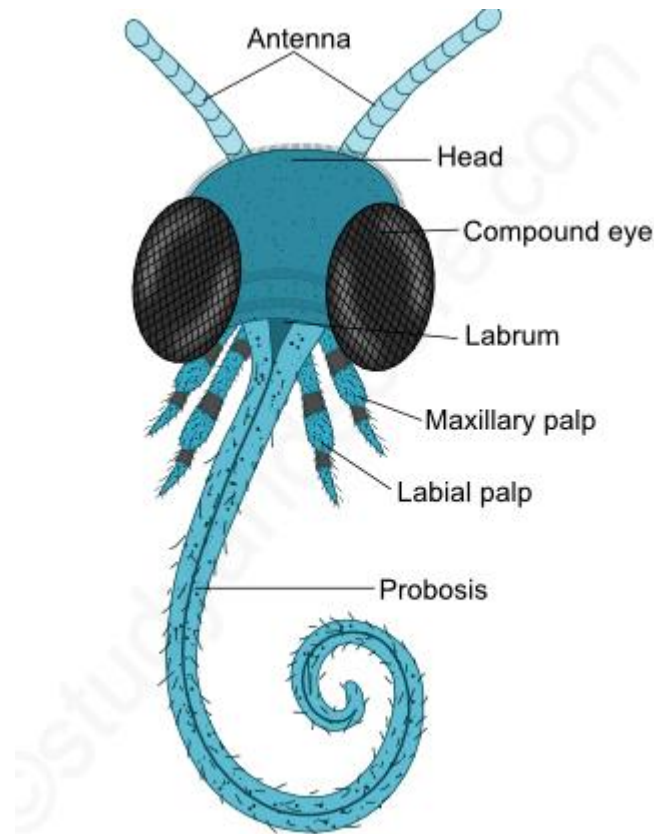
This type of mouth parts is adapted for piercing the tissues of animals and plants to suck blood and plant juice. Found in dipteran insects like mosquitoes and hemipteran insects like bugs, aphids.



Mouthparts of a mosquito

SIPHONING MOUTHPARTS

These types of mouth parts are adapted for sucking flower nectar and fruit juice, especially in the insect order Lepidoptera, to which all the butterflies and moths belong.



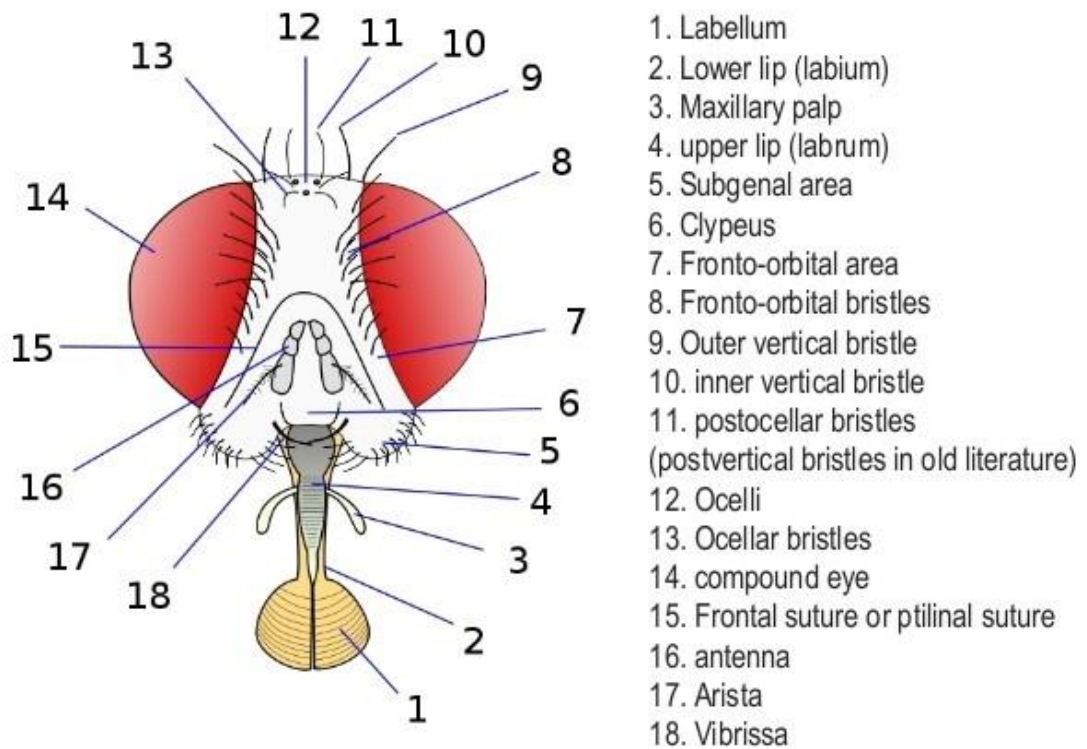
SPONGING MOUTHPARTS

This type of mouth parts are adapted for sucking up liquid or semiliquid food and found in houseflies and some other flies.

They consist of:

- | | |
|-----------------------|-----------------|
| 1. Labrum- epipharynx | 3. Labium |
| 2. Maxillae | 4. Hypo pharynx |

Mandibles are entirely absent. In fact, in this type of mouth parts, the labium, i.e., lower lip is well developed and modified to form a long, fleshy and retractile proboscis.

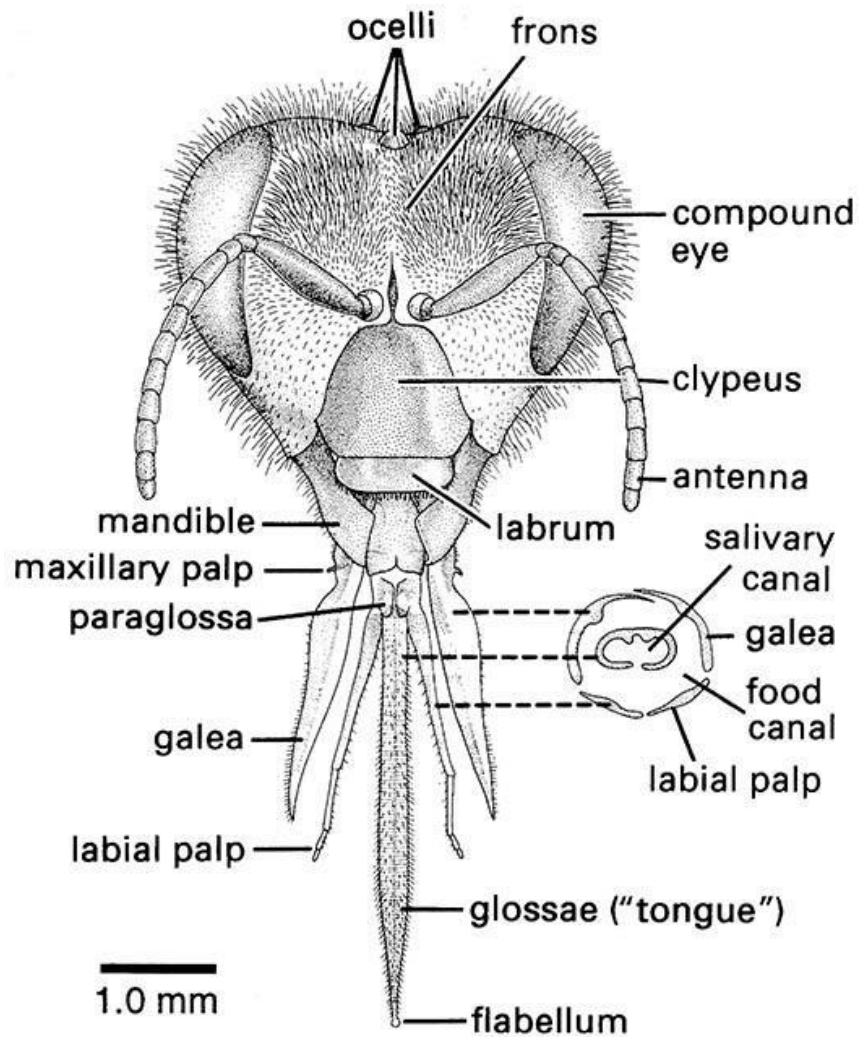


CHEWING AND LAPPING MOUTHPARTS

This type of mouth parts are modified for collecting the nectar and pollen from flowers and also for moulding the wax, as is found in honeybees, wasps, etc.

They consist of:

- | | |
|---------------|----------------------------|
| 1. Labrum | 4. First pair of maxillae |
| 2. Epipharynx | 5. Second pair of maxillae |
| 3. Mandibles | |



Activities and questions

1. Dissect the mouthparts of the given specimen and draw a complete diagram of it.
2. Identify the feeding habits of the given specimens by observing the mouthparts.
3. Draw and write special adaptations of the mouthparts of the following specimens.
 - a) Scarab beetle
 - b) Mosquito
 - c) Butterfly